

Advanced (Habanero)

Q1. What is the vertex form of $x^2 + 6x + 9$?

- (A) $(x + 3)^2$
- (B) $(x - 3)^2$
- (C) $x^2 + 6x + 8$
- (D) $x^2 - 6x - 9$
- (E) $x^2 + 3x + 2$

Q2. Which equation is in vertex form?

- (A) $x^2 + 2x + 1$
- (B) $x^2 - 4x - 3$
- (C) $(x + 1)^2$
- (D) $x^2 + 2x - 5$
- (E) $(x - 2)^2 + 1$

Q3. Complete the square for $x^2 + 4x$

- (A) $(x + 2)^2 - 4$
- (B) $(x - 2)^2 + 4$
- (C) $x^2 + 4x + 2$
- (D) $x^2 - 4x + 2$
- (E) $(x + 2)^2$

Q4. What is the value of a in the vertex form $(x - 2)^2 + 3$?

- (A) 1
- (B) 2
- (C) 3
- (D) 4
- (E) 0

Q5. Convert $x^2 - 6x + 5$ to vertex form

- (A) $(x - 3)^2 - 4$
- (B) $(x + 3)^2 + 4$
- (C) $(x - 3)^2 + 4$
- (D) $(x + 3)^2 - 4$
- (E) $x^2 - 6x - 5$

Q6. The vertex of the parabola $x^2 + 2x - 3$ is

- (A) $(-1, -4)$
- (B) $(-1, 0)$
- (C) $(-2, 0)$
- (D) $(-1, 4)$
- (E) $(1, -4)$

Q7. Which of the following is equivalent to $x^2 + 8x + 16$?

- (A) $(x + 4)^2 - 16$
- (B) $(x - 4)^2 + 16$
- (C) $(x + 4)^2$
- (D) $(x - 4)^2$
- (E) $x^2 + 4x + 2$

Q8. The equation $(x + 2)^2 - 1$ can be rewritten in standard form as

- (A) $x^2 + 4x + 4$
- (B) $x^2 + 4x + 3$
- (C) $x^2 + 4x - 1$
- (D) $x^2 - 4x - 1$
- (E) $x^2 - 4x + 1$

Open Ended Questions (FRQ)**Q9.** Convert $x^2 - 10x + 8$ to vertex form and identify the vertex.**Q10.** Complete the square for $x^2 + 12x$ and then rewrite the equation in vertex form.**Chef's Tips**

- Check calculations carefully
- Ensure proper use of algebraic notation
- Use the completed square form to identify the vertex